



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 2)

2017 HSC Course Assessment Task 1

Friday December 9, 2016

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided.

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Dr Jomaa

☐ 12M4B – Ms Lee

☐ 12M4C – Mr Lin

Marker's use only.

QUESTION	1-5	6	7	8	Total	%
MARKS	$\overline{5}$	$\overline{11}$	$\overline{14}$	$\overline{9}$	$\overline{39}$	

Section I

5 marks

Attempt Question 1 to 5

Mark your answers on the answer grid provided.

Questions	Marks
1. Which of the following is true for all complex numbers z ?	1
(A) $\operatorname{Im}(z) = \frac{z - \bar{z}}{2}$	
(B) $\operatorname{Im}(z) = \frac{z + \bar{z}}{2}$	
(C) $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$	
(D) $\operatorname{Im}(z) = \frac{z + \bar{z}}{2i}$	
2. Let the point R represent the complex number z on an Argand diagram. Which of the following describes the locus of R specified by $2 z = z + \bar{z} + 4$.	1
(A) Circle with centre $(0, 0)$ and radius 4.	
(B) Parabola with vertex $(-1, 0)$, axis $y = 0$	
(C) Parabola with vertex $(-1, 0)$, axis $x = 0$	
(D) Perpendicular bisector of $(0, 0)$ and $(0, 4)$.	
3. The polynomial $P(z) = z^3 + (1 + i)z^2 + (1 + i)z + 1$ has a real zero $z = -1$ and a complex zero $z = \alpha$. By considering the sum and the product of the roots, the third root is:	1
(A) $\frac{1}{\alpha}$	
(B) $\bar{\alpha}$	
(C) $-\alpha$	
(D) $1 - \alpha$	

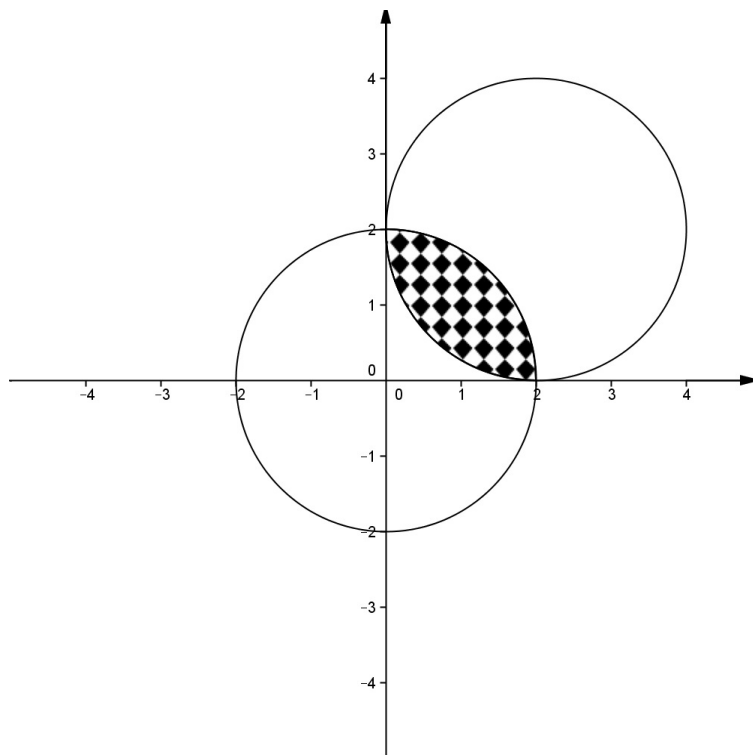
4. The shaded area in the Argand diagram below can be represented by the inequalities: 1

(A) $|z| \leq 4$ and $|z - 2 + 2i| \leq 4$

(B) $|z| \leq 2$ and $|z - 2 - 2i| \leq 2$

(C) $|z| \leq 4$ and $|z - 2 - 2i| \leq 4$

(D) $|z| \leq 2$ and $|z - 2 + 2i| \leq 2$



5. If ω is a non-real sixth root of -1 and ϕ is a non-real fifth root of 1, consider the following two statements: 1

(I) $1 - \omega + \omega^2 - \omega^3 + \omega^4 - \omega^5 = 0$.

(II) $1 + \phi + \phi^2 + \phi^3 + \phi^4 = 0$

(A) Both (I) and (II) are correct.

(B) Only (I) is correct.

(C) Only (II) is correct.

(D) Neither is correct.

Please Turn Over

Section II

34 marks

Attempt Questions 6 to 7

Write your answers in the writing booklets supplied. Additional writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (11 Marks)	Commence a NEW page.	Marks
(a) Let $z_1 = 1 - i$ and $z_2 = -1 + i\sqrt{3}$.		
i. Express $z = z_1 z_2$ in the form $a + ib$, where a and b are real.		2
ii. By expressing z_1 and z_2 in the modulus argument form, show that		2
$z = \sqrt{8} \operatorname{cis} \frac{5\pi}{12}.$		
iii. Hence, write down the exact values of $\cos \frac{5\pi}{12}$ and $\sin \frac{5\pi}{12}$.		2
(b) The complex number z is a solution of the equation $\sqrt{z} = \frac{4}{1+i} + 7 - 2i$. Express z in the form $a + ib$, where a and b are integers.		2
(c) If ω is a non-real cube root of unity, show the following:		
i.	$(1 + \omega)^3 = -1$	1
ii.	$(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega) = 3$	2

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Question 7 (14 Marks)

Commence a NEW page.

Marks

- (a) The polynomial $P(x) = x^4 + ax^2 + bx + 28$ has double root at $x = 2$. What are the values of a and b ? **2**
- (b) The equation $x^3 - x - 1 = 0$ has roots α , β and γ . Find an equation whose roots are $\frac{1+\alpha}{1-\alpha}$, $\frac{1+\beta}{1-\beta}$, $\frac{1+\gamma}{1-\gamma}$ in the form $Ax^3 + Bx^2 + Cx + D = 0$. **3**
- (c) The Polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ has roots $a + ib$ and $a + 2ib$, where a and b are real and $b \neq 0$.
- i. By evaluating a and b , find all roots of $P(x)$. **3**
- ii. Hence, or otherwise, find the quadratic polynomials with real coefficients that are factors of $P(x)$. **1**
- (d) i. Find real numbers a and b , such that **2**
- $$z^4 - z^3 + z^2 - z + 1 = (z^2 + az + 1)(z^2 + bz + 1).$$
- ii. Given that $z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$ is a solution of $z^4 - z^3 + z^2 - z + 1 = 0$, find the exact value of $\cos \frac{\pi}{5}$. **3**

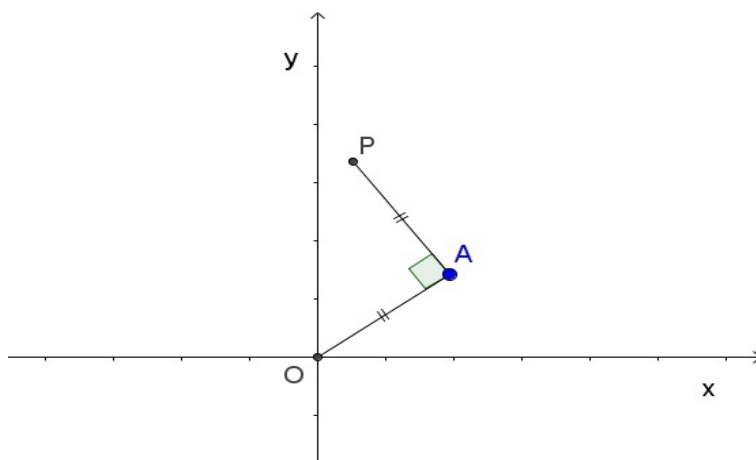
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Question 8 (9 Marks)

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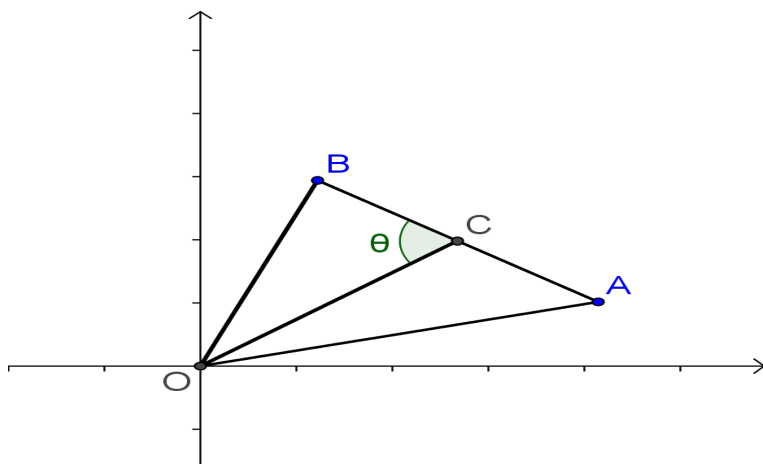
Marks

- (a) The point A in the complex plane corresponds to complex numbers z . The triangle OAP is a right-angled isosceles triangle.



- i. What complex number corresponds to the point P? 1
- ii. Let M is the midpoint of OP. What complex number corresponds to M? 1

(b)



A and B are the points which represent the complex numbers z and w respectively. Let C be the midpoint of AB. By taking $\angle BCO = \theta$ and the cosine rule, prove that 3

$$|z - w|^2 + |z + w|^2 = 2|z|^2 + 2|w|^2.$$

- (c) Sketch the region in the Argand diagram consisting of those points z for which 4
 $|z| > 1$ intersect with $\frac{\pi}{4} \leq \arg\left(\frac{z-1}{z+1}\right) \leq \pi$. Show all important features.

End of paper.

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Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

STUDENT NUMBER:

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1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

2017 Extension 2 Task 1 solutions

Section I:

1. C
2. B
3. A
4. B
5. C

Section II:

6.

a) $z_1 = 1 - i, z_2 = -1 + i\sqrt{3}$

i. $z = z_1 z_2 = (1 - i)(-1 + i\sqrt{3}) = -1 + \sqrt{3} + i(\sqrt{3} + 1)$

ii. $z_1 = \sqrt{2}\text{cis}\left(\frac{-\pi}{4}\right), z_2 = 2\text{cis}\left(\frac{2\pi}{3}\right)$

$$z = z_1 z_2 = 2\sqrt{2} \text{cis}\left(\frac{-\pi}{4} + \frac{2\pi}{3}\right) = \sqrt{8} \text{cis}\left(\frac{5\pi}{12}\right).$$

iii. $-1 + \sqrt{3} + i(\sqrt{3} + 1) = \sqrt{8} \text{cis}\left(\frac{5\pi}{12}\right)$

$$\cos \frac{5\pi}{12} = \frac{-1 + \sqrt{3}}{\sqrt{8}}, \text{ and } \sin \frac{5\pi}{12} = \frac{1 + \sqrt{3}}{\sqrt{8}}.$$

b) $\sqrt{z} = \frac{4}{1+i} + 7 - 2i = \frac{4(1-i)}{(1+i)(1-i)} + 7 - 2i = 2(1-i) + 7 - 2i = 9 - 4i.$

$$z = (9 - 4i)^2 = 81 - 72i + 16i^2 = 81 - 16 - 72i = 65 - 72i.$$

c) $\omega^3 = 1 \therefore (\omega - 1)(\omega^2 + \omega + 1) = 0 \therefore 1 + \omega + \omega^2 = 0.$ Since ω is a non-real cube root of unity.

i. $(1 + \omega)^3 = 1 + 3\omega + 3\omega^2 + \omega^3 = 3 + 3\omega + 3\omega^2 + \omega^3 - 2 = 3(1 + \omega + \omega^2) + \omega^3 - 2 = 3 \times 0 + 1 - 2 = -1.$ Since $\omega^3 = 1.$

ii. $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega) = (\omega + 2\omega^2)(\omega^2 + 2\omega) = (\omega^2 - 1)(\omega - 1) = 1 - (\omega + \omega^2) + 1 = 2 - (-1) = 3.$

7.

a) $P(x) = x^4 + ax^2 + bx + 28$ has double root at $x = 2.$

$$\therefore P(2) = 0 \text{ and } P'(2) = 0.$$

$$P'(x) = 4x^3 + 2ax + b$$

$$P(2) = 0 \therefore 16 + 4a + 2b + 28 = 0 \text{ and}$$

$$P'(2) = 0 \therefore 32 + 4a + b = 0.$$

$$\text{Now } 2a + b = -22 \text{ and } 4a + b = -32$$

$$\text{Solving simultaneously } a = -5, b = -12.$$

b) $x^3 - x - 1 = 0$ has roots α, β and $\gamma.$ Let $y = \frac{1+\alpha}{1-\alpha}$ and makes α the subject.

$$\alpha = \frac{y-1}{y+1}. \text{ Since } \alpha \text{ is a root of } x^3 - x - 1 = 0 \therefore \left(\frac{y-1}{y+1}\right)^3 - \left(\frac{y-1}{y+1}\right) - 1 = 0$$

$$(y-1)^3 - (y-1)(y+1)^2 - (y+1)^3 = 0$$

$$y^3 - 3y^2 + 3y - 1 - (y-1)(y^2 + 2y + 1) - (y^3 + 3y^2 + 3y + 1) = 0$$

$$-6y^2 - 2 - (y^3 + 2y^2 + y - y^2 - 2y - 1) = 0$$

$$-y^3 - 7y^2 + y - 1 = 0$$

Hence $x^3 + 7x^2 - x + 1 = 0$ has roots $\frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}$ and $\frac{1+\gamma}{1-\gamma}$.

c)

$$P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10 \text{ has roots } a + ib \text{ and } a + 2ib.$$

i. By the complex conjugate theorem $a - ib$ and $a - 2ib$ also are roots.

$$\text{Sum of the roots} = 4 = 4a \therefore a = 1.$$

$$\text{Product of roots} = 10 = (a + ib)(a - ib)(a + 2ib)(a - 2ib) = (a^2 + b^2)(a^2 + (2b)^2)$$

$$\text{Now using } a = 1 \therefore (b^2 + 1)(4b^2 + 1) = 10$$

$$4b^4 + 5b^2 - 9 = 0$$

$$\therefore b^2 = 1 \text{ or } b^2 = \frac{-9}{4}$$

$$\therefore b^2 = 1 \text{ since } b \text{ is real. and } b = \pm 1$$

So the roots are $1 + i, 1 - i, 1 + 2i$ and $1 - 2i$.

ii. The quadratic polynomials with real coefficients are

$$x^2 - 2ax + a^2 + b^2 \text{ and } x^2 - 2ax + a^2 + 4b^2$$

$$\text{Hence } x^4 - 4x^3 + 11x^2 - 14x + 10 = (x^2 - 2x + 2)(x^2 - 2x + 5)$$

d)

$$\text{i. } z^4 - z^3 + z^2 - z + 1 = (z^2 + az + 1)(z^2 + bz + 1)$$

Expand the right hand side, we obtain:

$$z^4 - z^3 + z^2 - z + 1 = z^4 + (a + b)z^3 + (2 + ab)z^2 + (a + b)z + 1$$

Hence $a + b = -1$ and $2 + ab = 1$. Now $a + b = -1$ and $ab = -1$.

Solving simultaneously, we obtain $a^2 + a - 1 = 0$ where $a = \frac{-1 \pm \sqrt{5}}{2}$.

$$\text{So } a = \frac{-1 + \sqrt{5}}{2} \text{ and } b = \frac{-1 - \sqrt{5}}{2}.$$

ii. $z^5 = -1$. Let $z = r(\cos \theta + i \sin \theta)$, applying De Moivre's theorem, we obtain:

$$r^5 \text{cis } 5\theta = \text{cis } \pi, \text{ so } r = 1 \text{ and } 5\theta = \pi + 2k\pi = (2k + 1)\pi$$

$$\text{We need } -\pi < \theta \leq \pi. \quad -\pi < \frac{(2k+1)\pi}{5} \leq \pi \text{ and } -3 < k \leq 2.$$

Hence $k = -2, -1, 0, 1, 2$. The roots are:

$$\text{cis } \frac{-3\pi}{5}, \text{cis } \frac{-\pi}{5}, \text{cis } \frac{\pi}{5}, \text{cis } \frac{3\pi}{5}, \text{cis } \frac{5\pi}{5} = -1$$

Now $z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$ corresponds to $k = 0$, and its conjugate corresponds to

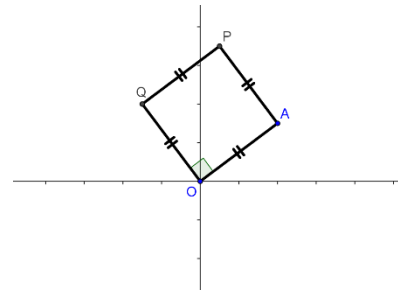
$$k = -1. \text{ Sum of the roots} = \text{cis } \frac{\pi}{5} + \text{cis } \frac{-\pi}{5} = 2 \cos \frac{\pi}{5} = -b, (a > 0).$$

$$\text{Hence } \cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4}.$$

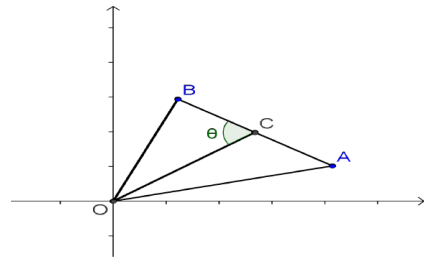
8.

a)

- i. The point A corresponds to the complex number z . Q corresponds to $\overrightarrow{AP} = iz$.
 $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = z + iz = (1+i)z$
Hence P corresponds to the complex number $(1+i)z$.
- ii. M is the midpoint of OP and the corresponding complex number is $\left(\frac{1+i}{2}\right)z$.



- b) In triangle OBC,
 $OB^2 = OC^2 + BC^2 - 2 OC BC \cos \theta$ (1)
Similarly in triangle OAC,
 $OA^2 = OC^2 + AC^2 - 2 OC AC \cos(180 - \theta)$ (2)
But $\cos(180 - \theta) = -\cos \theta$ and
 $AC = BC$ (C is the midpoint of AB)



Now (1)+(2) gives:

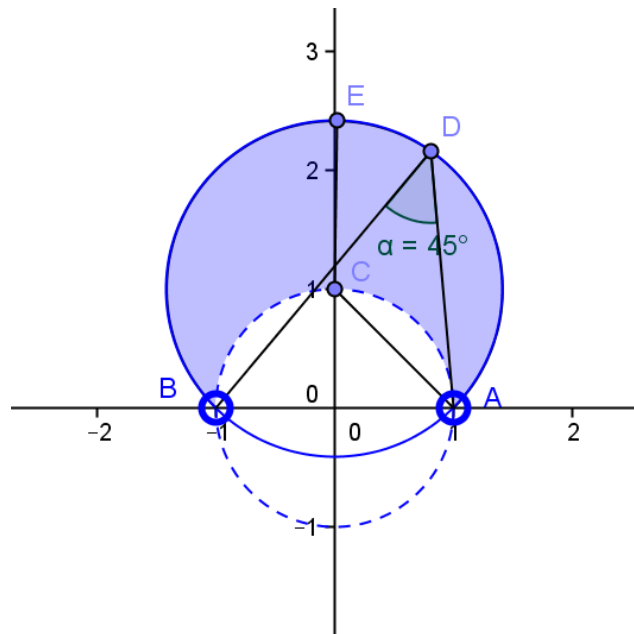
$$OA^2 + OB^2 = 2(OC^2 + BC^2)$$

$$OA = |z|, OB = |\omega|, OC = \frac{|z + \omega|}{2} \text{ and } BC = \frac{|z - \omega|}{2}$$

$$|z|^2 + |\omega|^2 = 2\left(\frac{|z + \omega|^2}{4} + \frac{|z - \omega|^2}{4}\right)$$

$$\text{Hence } 2(|z|^2 + |\omega|^2) = |z + \omega|^2 + |z - \omega|^2.$$

c)



C(0,1) and the radius is $\sqrt{2}$.

